VII SEMESTER (PROFESSIONAL ELECTIVE-IV)	L	T	P	C
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20AM7E05 - VIDEO ANALYTICS				

COURSE OBJECTIVES:
The main objectives of the course is to make student understand the need for video Analytics, the basic configuration of video analytics, the functional blocks of a video analytic system and to get exposed to the various applications of video analytics

COURSE OUTCOMES:
At the end of the course, the student should be able to:
CO1: Design video analytic algorithms for security applications
CO2: Design video analytic algorithms for business intelligence
CO3: Design custom made video analytics system for the given target application
CO4: Understanding of the principles and concepts underlying video analytics**.**
CO5: Understand how preprocessing techniques can improve the accuracy and reliability of video analytics algorithms

UNIT -I:
Video Analytic Components: Need for Video Analytics, Overview of video Analytics, Foreground extraction, Feature extraction, classifier, Preprocessing, edge detection, smoothening, Feature space-PCA-FLD-SIFT features

UNIT -II:
Foreground Extraction: Background estimation, Averaging, Gaussian Mixture Model, Optical Flow based, Image Segmentation, Region growing, Region splitting, Morphological operations, erosion, Dilation, Tracking in a multiple camera environment

UNIT -III:
Classifiers: Neural networks (back propagation), Deep learning networks, Fuzzy Classifier, Bayesian classifier, HMM based classifier

UNIT -IV:
Video Analytics for Security: Abandoned object detection, human behavioral analysis, human action recognition, perimeter security, crowd analysis and prediction of crowd congestion

UNIT -V:
Video Analytics For Business Intelligence & Traffic Monitiring And Assistance: Customer behavior analysis, people counting, Traffic rule violation detection, traffic congestion identification for route planning, driver assistance, lane change warning

Text Books:
1. Graeme A. Jones, Nikos Paragios, Carlo S. Regazzoni, “Video-Based Surveillance Systems: Computer Vision and Distributed Processing”, Kluwer academic publisher, 2001
2. Nilanjan Dey, Amira Ashour and Suvojit Acharjee, “Applied Video Processing in Surveillance and Monitoring Systems”, (IGI global) 2016

Reference Books:
1. Zhihao Chen, Ye Yang, Jingyu Xue, Liping Ye, Feng Guo, “The Next Generation of Video Surveillance and Video Analytics: The Unified Intelligent Video Analytics Suite”, CreateSpace Independent Publishing Platform, 2014
2. Caifeng Shan, Fatih Porikli, Tao Xiang, Shaogang Gong, “Video Analytics for Business Intelligence”, Springer, 2012.